



COMPANY PROFILE 2026

Shaping sustainable futures through **technology, innovation, and community**

4D Climate Solutions empowers communities across Southern Africa to actively engage in climate solutions — combining secure digital systems, applied AI, and deep community engagement to build resilience and improve livelihoods.



A Lesotho-grown climate and technology firm

4D Climate Solutions (4D) was founded in 2018 by three climate action enthusiasts in Lesotho, driven by a mission to empower communities to actively engage in solutions to combat climate change and enhance their resilience. What sets 4D apart is **how** we deliver: cutting-edge technology combined with deep community engagement, ensuring sustainable and lasting impact.

Over seven years, technology has become the backbone of our work. We have helped government and development programs move from paper registers, spreadsheets, and manual verification to secure, auditable, real-time digital systems — designed for low-bandwidth environments, local languages, and field conditions. Today 4D builds and operates live production platforms for public-sector programs, including the compensation information system serving the Lowlands Water Development Project.

4D AT A GLANCE

7+ YEARS OF IMPACT	50+ PROJECTS DELIVERED	8 DISTRICTS COVERED	5 COUNTRIES OF EXPERIENCE
12+ DIGITAL APPLICATIONS BUILT	45,000+ PLATFORM USERS (PILOT)	M20+ CONTRACT VALUE	20+ PARTNER ORGANIZATIONS

OUR MISSION

To empower communities to actively engage in solutions to combat climate change and enhance their resilience — using technology that puts reliable information in the hands of government, field teams, and citizens.

OUR DIFFERENCE

We do not only build systems — we run the community mobilization, training, and change management that make them stick. Every 4D platform ships with capacity building, data governance, and a sustainability plan.

Technology & innovation capabilities

Six capability pillars cover the full lifecycle of a modern public program — from system design and data management to field rollout, training, and long-term operation.

01 Government information systems & registries

Design and development of secure management information systems for public programs: beneficiary and asset registries, compensation management, grievance redress, monitoring & evaluation, and administrative workflows. Systems include role-based access control, approval chains, full audit trails, and print-ready official forms.

02 Digital field data collection

Offline-first mobile and web applications for surveys, censuses, verification exercises, and community consultations (pitsos, FGDs, KIIs) — with GPS capture, photo and document capture, and real-time dashboards for supervisors. Built to work on 2G/3G networks and sync automatically when connectivity returns.

03 AI & smart data systems

AI-powered decision support and advisory platforms through our dedicated 4D AI division (ai.4dcs.co.za): retrieval-augmented (RAG) advisory systems, AI vision for real-time monitoring, and predictive analytics — engineered for low-bandwidth environments, local languages, and African realities, using Anthropic Claude AI.

04 Secure cloud infrastructure & data protection

Cloud-hosted architectures with authenticated access, database row-level security, encrypted document storage, allow-listed user registration, and immutable change logs — so every edit to a public record is attributable and reviewable.

05 GIS, remote sensing & IoT

Geospatial analysis, remote-sensing assessments, and IoT sensor networks — from groundwater aquifer classification (with the University of Virginia) to solar-powered smart irrigation with live soil-moisture telemetry.

06 Capacity building & technology adoption

Tailor-made training for government staff, field teams, and communities; formation of community structures; user onboarding, help materials, and phased handover — ensuring systems are owned and sustained locally after project close.

CIMS — Compensation Information Management System

End-to-end management of asset registration, verification, valuation, and compensation tracking for Project Affected Persons (PAPs) under the **Lesotho Lowlands Water Development Project Phase III (LLWDP III)** — replacing paper files and manual reconciliation with a single, secure, auditable source of truth.

Client context: LLWDP III (African Development Bank–financed)

Status: Live — cims.4dcs.co.za

Developed & operated by: 4D Climate Solutions

1,388

PAP RECORDS
MANAGED

94%

RECORDS FIELD-
VERIFIED

25

ROUTES (17 RURAL · 8
URBAN)

16

COMMUNAL ASSET
CATEGORIES

WHAT CIMS DELIVERS

- **Household & asset registry.** Beneficiaries, co-owners, land parcels, structures, and communal assets in one linked database.
- **Verification workflow.** Print-ready two-page verification forms with photos and ID documents; field completion writes straight back to the database.
- **Compensation valuation.** A central Rates Master governs land and asset rates; compensation auto-calculates (area × rate + disturbance) — one source of truth, no spreadsheet drift.
- **Payment tracking.** Payment status and payment documents per PAP, with live dashboard counters for management oversight.
- **Document management.** IDs, photos, and payment evidence uploaded from the field to encrypted cloud storage, linked to each record.

GOVERNANCE & DATA INTEGRITY

- **Role-based access.** Distinct tiers for administrators, community liaison and resettlement officers, field surveyors, and view-only client users.
- **Approval chain.** Ordinary edits flow through an edit-request and approval workflow rather than direct changes to official records.
- **Live audit trail.** Every change is logged and visible in a real-time Change History on each record.
- **Row-level security.** Database-enforced permissions on every table; registration restricted to an authorized email allow-list.
- **De-duplication.** Guided merge workflows resolve duplicate PAP files — including multi-asset households — with a reviewable result before commit.

TECHNOLOGY

React single-page application · PostgreSQL (Supabase) with row-level security and realtime subscriptions · Express API · Cloudflare R2 encrypted object storage · continuous deployment on Render. Delivered with user training, operational handover notes, and live production support.

Platforms built and operated by 4D

Twelve-plus custom applications delivered for government programs, development partners, and local enterprises — designed offline-first for low-bandwidth environments.

Lipalo — AI Agricultural Advisory

LIVE

Lesotho's AI-powered agricultural advisory system (v6.8.0), bridging a 1:1,500 extension-officer-to-farmer gap. Built on Anthropic Claude AI with 101 agronomy knowledge chunks and 40+ API endpoints.

- Real-time weather alerts with farming impact analysis
- FAO-based crop calendars for 4 agro-ecological zones
- LESIS-based soil analysis with crop suitability ratings
- FAO-56 irrigation scheduling — 8 crops, all 10 districts

LLWDP II Monitoring App

LIVE

Field data collection for community mobilization under the Lowlands Water Development Project Phase II — capturing consultations end-to-end.

- Offline-first PWA architecture for remote districts
- Pitsos, FGDs, KIIs, and baseline surveys
- GPS location capture on every activity
- Real-time administrative dashboard

Letlotlo (Gooi Gooi Savings)

LIVE

Financial inclusion platform for savings groups, burial societies, street vendors, and factory workers — with Business Apps, KTB, and MOB.

- 45,000+ users across 2,600+ groups (pilot)
- VSLA, burial society & group management
- Mobile money integration & loan access facilitation

Kopanya Fresh

LIVE

Agricultural marketplace connecting smallholder farmers with urban buyers.

- Three-tier crop pipeline tracking — Ready Now / Harvesting Soon / Coming Soon
- Buyer, Farmer & Agent portals
- Order management & delivery coordination

Rekisa POS

LIVE

Multi-tenant point-of-sale system for Lesotho businesses.

- Inventory management, sales tracking & reports
- Multi-branch support
- Offline transaction sync

GrowSmart IoT Irrigation

IN DEVELOPMENT

Solar-powered, IoT-based smart irrigation implemented with LECSA in Morija.

- Live soil-moisture telemetry & weather integration
- Remote pump control from a mobile app
- Predictive analytics & personalized crop advice

Research-grade AI, built and tested in Lesotho

RESEARCH COLLABORATION · UNIVERSITY OF VIRGINIA

AI for Agricultural Resilience — UVA × 4D Decision Support System

With faculty at the University of Virginia and the National University of Lesotho, 4D is piloting its AI-powered Decision Support System (DSS): a retrieval-augmented (RAG) assistant that turns ReNOKA's 500-page Compendium of soil and water conservation measures into real-time, actionable guidance for extension agents and farmers — integrated with Land Cover Atlas data and continuously improved through expert-in-the-loop feedback (RLHF), with Sesotho-language support planned.

Design: Randomized evaluation with 17 extension agents serving 340 farmers**Horizon:** Two planting seasons, 2026–2027

YeboSell

Digital commerce platform helping small enterprises and traders sell, track, and grow their business online.

Skolo

Digital platform for schools — bringing administration, learning resources, and communication into one accessible system.

Tafela

Digital ordering and hospitality platform connecting customers with local food businesses.

RxPlan

AI-based prescription and medication planning support for health facilities and pharmacies.

ONE ENGINEERING PRACTICE, MANY DOMAINS

From agriculture and water to finance, health, education, and commerce, every 4D platform shares the same foundations: offline-first design, secure cloud infrastructure, role-based access, and AI applied where it multiplies human capacity.

Delivering for government & development partners

Selected current and completed assignments — with the technology and innovation component of each engagement highlighted.

PROJECT	CLIENT / PARTNER	TECHNOLOGY & INNOVATION
LLWDP III — Asset Verification & Compensation GOV 	African Development Bank—financed; community mobilization, RAP implementation & grievance redress in Botha-Bothe & Leribe	CIMS — live registry of 1,388 PAPs with verification, valuation & payment tracking (see page 4)
LLWDP II Community Mobilization Study GOV Oct 2025 – Feb 2026	Ministry of Natural Resources · JV with Khanyisa Projects	Digital monitoring system for all field data collection — pitsos, FGDs & surveys captured on an offline-first app with live dashboards
SADP II National Service Provider — Southern Region GOV May 2025 – Jun 2026	Ministry of Agriculture, Food Security & Nutrition · World Bank (IDA 64220)	Digital administration of matching-grant beneficiaries and field service provider management
Lipalo — AI Agricultural Extension Ongoing	University of Virginia · National University of Lesotho	National-scale AI advisory platform — weather, crop calendars, soils & irrigation across all 10 districts
AI for Agricultural Resilience (DSS) 2026 – 2027	University of Virginia · National University of Lesotho	RAG-based AI decision support on the ReNOKA Compendium, evaluated with 17 extension agents and 340 farmers
LHWP Climate Change Vulnerability Assessment Aug 2022 – Dec 2023	Lesotho Highlands Development Authority	Data-driven vulnerability assessment and adaptation/mitigation strategy for LHWP Phases I & II
Agricultural Insurance Feasibility Study Feb – Dec 2020	IFC (World Bank Group) · RMSI India	Market and data analysis for smallholder agricultural insurance across six value chains
Value of Water Study Ongoing	University of Virginia	Economic valuation of water resources in highland communities, incl. community-led digital soil & water monitoring
Gooi Gooi Savings Solutions Ongoing	Business Apps · KTB · MOB	Digital savings system promoting financial inclusion for street vendors and factory workers
GrowSmart Solar Irrigation Ongoing	LECSA (Morija)	IoT sensor network, predictive analytics & mobile-app control for climate-smart irrigation

Our approach to technology & innovation

Principles proven across our live systems — the same standards we bring to every government and development engagement.

BUILT FOR THE FIELD, NOT JUST THE OFFICE

Offline-first applications that keep working on 2G/3G networks and sync when connectivity returns; GPS, photo, and document capture at the point of service; print-ready official forms where paper is still required.

SECURITY & ACCOUNTABILITY BY DESIGN

Role-based access, database row-level security, allow-listed registration, approval workflows for changes to official records, and immutable audit logs — every edit attributable, every record reviewable.

ONE SOURCE OF TRUTH

Central master data (rates, categories, registries) drives every calculation and report, eliminating spreadsheet drift and reconciliation disputes between departments, contractors, and communities.

AI WHERE IT MULTIPLIES CAPACITY

Applied AI through our 4D AI division — advisory chat, AI vision monitoring, and predictive analytics — designed for local languages, low bandwidth, and responsible public-sector use.

ADOPTION, NOT JUST DEPLOYMENT

Community sensitization, staff training, user onboarding, and phased handover with documentation — the change management that determines whether new technology sticks.

COST-EFFECTIVE & LOCALLY SUSTAINED

Lean cloud architectures with modest running costs; a flexible staffing model that mobilizes specialist expertise per engagement; systems documented for long-term local operation.

TECHNOLOGY & DELIVERY PARTNERS

TECHNOLOGY

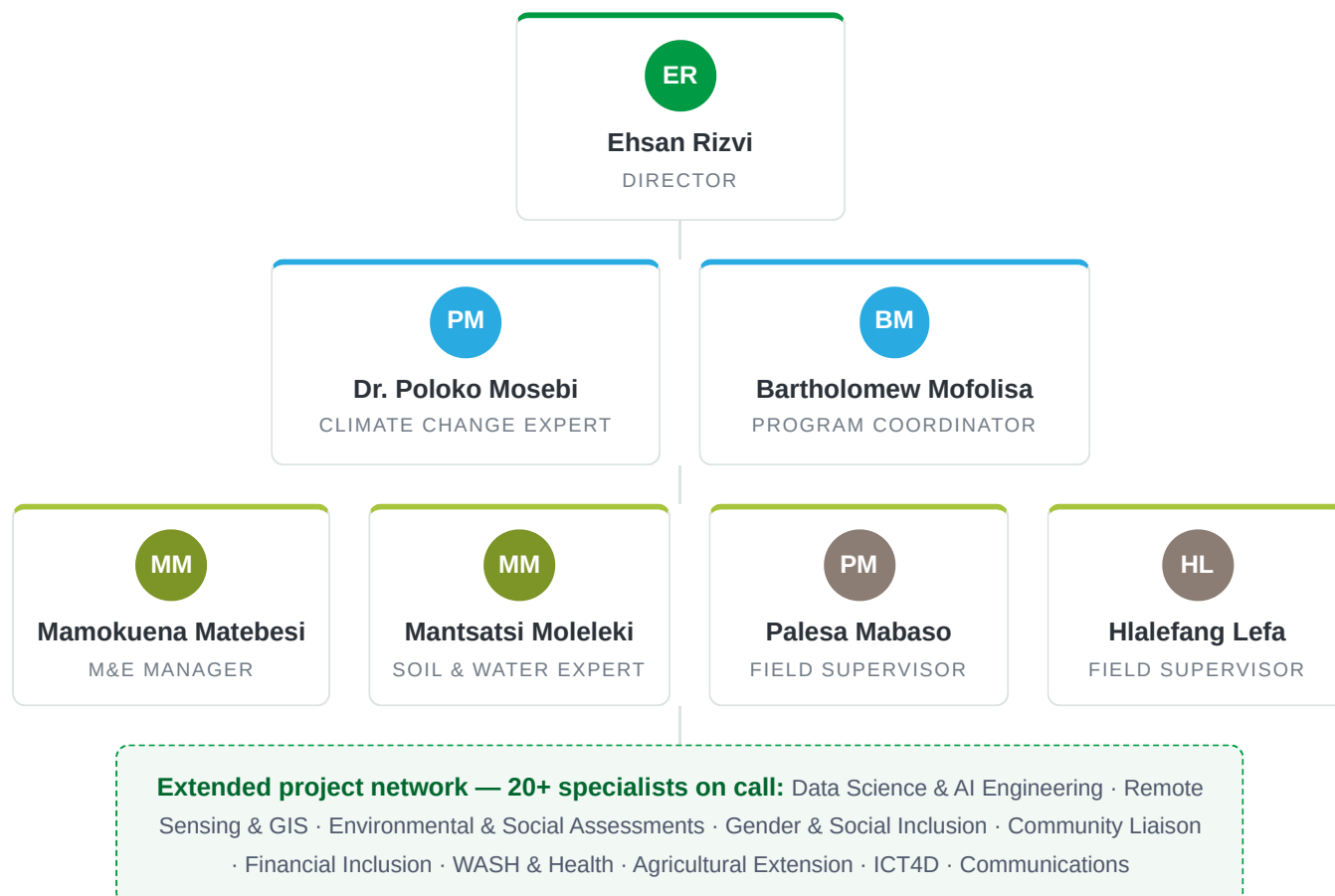
- Anthropic — Claude AI engine for Lipalo & advisory systems
- Supabase — cloud database & authentication infrastructure
- Cloudflare — encrypted object storage & delivery
- Render — cloud hosting & continuous deployment
- SunMoksha / G8Way Technologies — IoT & hardware

DELIVERY & RESEARCH

- Khanyisa Projects — joint venture partner
- University of Virginia (USA) · Bangor University (UK)
- National University of Lesotho (NUL)
- RMSI India Pvt. Ltd.
- Business Apps (Pty) Ltd. / OOWEBOO Web Engineers

Leadership & team

A lean core team, supported by a dynamic network of 20+ specialized project staff mobilized per engagement.



DIRECTOR · Ehsan Rizvi

A development professional with over 20 years in senior management roles across climate change, food security, natural resource and watershed management, agricultural livelihoods, social protection, and cash-based emergency response — with programs in India, Afghanistan, Sri Lanka, Tanzania, and Lesotho. Ehsan leads the design and implementation of 4D's innovative projects and digital systems, has mobilized significant resources for land restoration, implemented community-led soil and water monitoring systems, and collaborated with the University of Virginia on groundwater aquifer classification. He holds a PG Diploma in Food Security & Climate Change from Bangor University, UK, and a B.Tech in Civil Engineering from IIT Kanpur.

CLIMATE CHANGE EXPERT · Dr. Poloko Mosebi

PhD in Plant Science and MSc in Pasture Science (University of Pretoria). Leads key initiatives including LLWDSP III resettlement efforts and the GrowSmart initiative, integrating advanced technology with sustainable agricultural practices.

PROGRAM COORDINATOR · Bartholomew Mofolisa

Over 20 years of experience in livelihood projects, market system development, and human-centered supply chains across Lesotho.

Clients, partners & registration

We collaborate with leading development organizations, government institutions, and research partners.

GOVERNMENT & PUBLIC SECTOR

- Ministry of Natural Resources, Lesotho
- Ministry of Agriculture, Food Security & Nutrition, Lesotho
- Lesotho Highlands Development Authority (LHDA)
- Government of Lesotho

RESEARCH & ACADEMIC INSTITUTIONS

- National University of Lesotho (NUL)
- University of Virginia (USA)
- Bangor University (UK)
- RMSI India Pvt. Ltd.

INTERNATIONAL DEVELOPMENT PARTNERS

- World Bank / International Finance Corporation (IFC)
- African Development Bank (AfDB)
- Millennium Challenge Corporation (MCC)
- University of Virginia (UVA)

LOCAL & REGIONAL PARTNERS

- Khanyisa Projects (Joint Venture Partner)
- Caritas Lesotho · Care for Basotho Association
- Khathang Tema Batšukuli (KTB) / Mme-O-Bohlokoa (MOB)
- LECSA — Lesotho Evangelical Church in Southern Africa
- Business Apps (Pty) Ltd. / OOWEBOO Web Engineers

COMPANY OWNERSHIP & REGISTRATION

4D Climate Solutions (Pty) Ltd. was officially incorporated in May 2018. It operates as a single-shareholding company registered under License no. TR55456 with the Ministry of Trade and Industry, Cooperative and Marketing. Registration falls under the Industrial License Act of 1969 and Trading Enterprise Act of 1993.

WHERE WE WORK

Headquartered in Maseru, Lesotho, with project experience across 8 districts of Lesotho and programs delivered in 5 countries — India, Afghanistan, Sri Lanka, Tanzania, and Lesotho.



Let's shape sustainable futures — together.

From CIMS to Lipalo, 4D has shown that world-class digital public infrastructure can be designed, built, and operated from Maseru. We would welcome the opportunity to demonstrate our live systems and discuss how our experience can serve your program.

VISIT US

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